

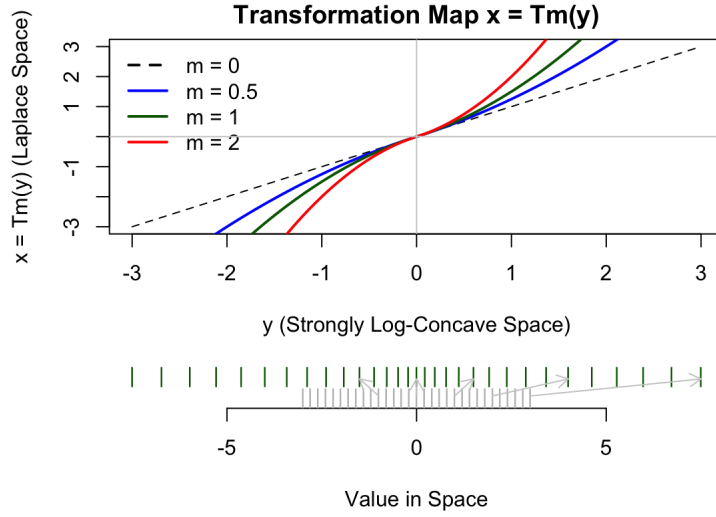
Laplace Random Variables by ‘Flattening’ of Strong-Log-Concavity

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Let $m > 0$, and define the bijection $T_m : \mathbf{R} \rightarrow \mathbf{R}$ by

$$T_m(y) = y + \frac{m}{2} \cdot y \cdot |y|.$$

Figure 1: Visualising this transformation.



Suppose that X is drawn from a standard Laplace distribution on $\mathbf{R}$, i.e. $X \sim Q$, where Q has density

$$q(x) = \frac{1}{2} \cdot \exp(-|x|),$$

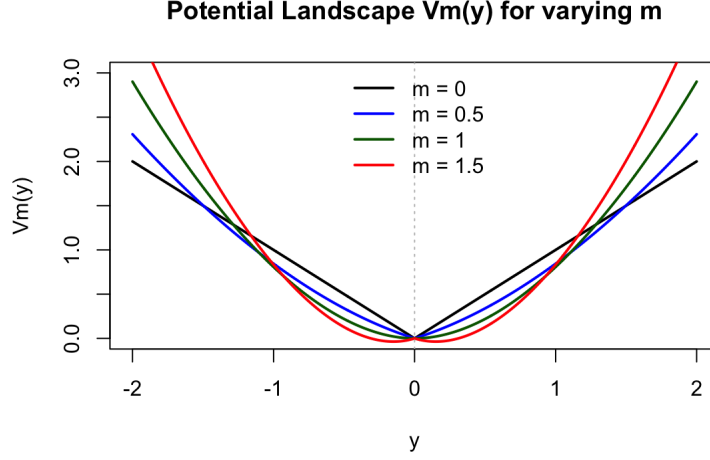
and define the random variable Y_m by $X = T_m(Y_m)$. Write P_m for the law of Y_m , and use the change-of-variable formula to compute (noting that $T'_m(y) = 1 + m \cdot |y|$) that P_m has density

$$p_m(y) = \frac{1}{2} \cdot \exp\left(-\left\{|y| + \frac{m}{2} \cdot |y|^2 - \log(1 + m \cdot |y|)\right\}\right).$$

Now, rewrite the negative log-density of p_m (up to a constant shift) as

$$V_m(y) = |y| + \frac{m}{2} \cdot |y|^2 - \log(1 + m \cdot |y|).$$

Figure 2: Examining the shape of the induced potential.



The non-smoothness of V_m makes checking its convexity slightly harder than usual, since the second derivative of V_m becomes a slightly strange beast, heuristically taking the form

$$V_m''(y) = \delta(y) + m \cdot \left\{ 1 - \frac{m}{(1 + m \cdot |y|)^2} \right\},$$

which would suggest convexity for $m \in [0, 1]$.

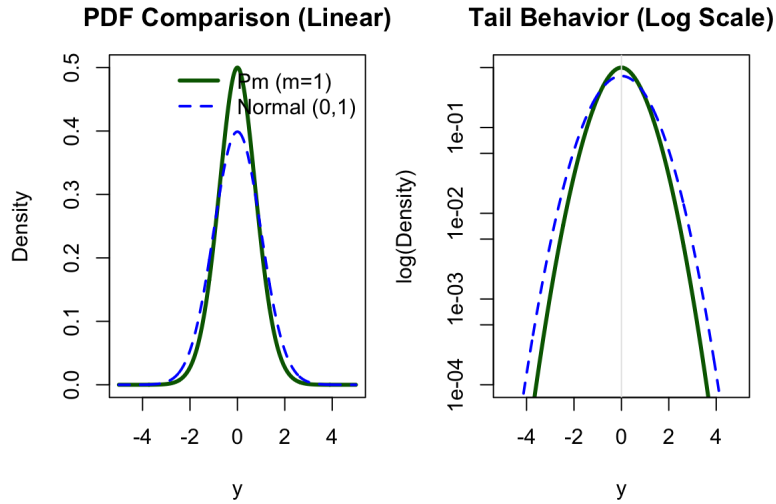
Nevertheless, examining the convexity of V_m is indeed rather tractable. Recalling that $z \mapsto z - \log(1 + z)$ is increasing and convex on $\mathbf{R}_+$, check that $w \mapsto w - \log(1 + |w|)$ is convex on all of $\mathbf{R}$, and re-express V_m as

$$V_m(y) = (1 - m) \cdot |y| + \frac{m}{2} \cdot |y|^2 + m \cdot |y| - \log(1 + m \cdot |y|).$$

One thus sees that for $m \in [0, 1]$, V_m remains convex, and for $m \in (0, 1]$, it is even strongly convex, with curvature parameter equal to m . At the limit $m = 1$, one specifically has

$$V_m(y) = \frac{1}{2} \cdot |y|^2 + \{|y| - \log(1 + |y|)\}.$$

Figure 3: Approximately normal behaviour at $m = 1$.



For $m > 1$, one checks that for $y \sim 0$, it holds that

$$V_m(y) = (1 - m) \cdot |y| + \Theta(|y|^2),$$

which is even locally concave.

Isoperimetrically speaking, one can also make some calculations. Using that

$$T_m^{-1}(x) = \text{sign}(x) \cdot \frac{\sqrt{1 + 2 \cdot m \cdot |x|} - 1}{m}$$

and calling on old results of Bobkov-Houdré, one finds that for $m \in (0, 1]$, the profile takes the form

$$\iota_m(p) = \min\{p, 1 - p\} \cdot \left\{ 1 + m \cdot \left| T_m^{-1} \left(\log \left(\frac{1}{2 \cdot \min\{p, 1 - p\}} \right) \right) \right| \right\},$$

which can be simplified to

$$\begin{aligned} \iota_m(p) &= \min\{p, 1 - p\} \cdot \sqrt{1 + 2 \cdot m \cdot \log \left(\frac{1}{2 \cdot \min\{p, 1 - p\}} \right)} \\ &\underset{p \rightarrow 0^+}{\asymp} \sqrt{2 \cdot m} \cdot \min\{p, 1 - p\} \cdot \sqrt{\log \left(\frac{1}{2 \cdot \min\{p, 1 - p\}} \right)}. \end{aligned}$$

This expression invites a natural comparison to Gaussian-like isoperimetry (which is no great surprise).

Figure 4: Near-Gaussian isoperimetry.

